

CSE 416 Knicks' Voting Equipment Quality Score Calculation

Overview

This document explains how the Voting Equipment Quality Score is calculated. The score ranges from 0 (low-performing, obsolete, insecure), to 1 (high-performing, modern, secure). The score aggregates several subscores: Age, OS, Certification, Scan Performance, Error Rate, Reliability, and Security Risk.

0. Discontinued Over Everything

If a device is discontinued, it will automatically receive a score of 0. The reason behind this is we don't think an actively used device should receive a lower score than a device that is not trusted enough to be used at all

1. Age Score

Age Score reflects how old a device is relative to a 20-year usable lifespan.

Formula:

$$\text{Age} = \text{CurrentYear} - \text{FirstManufacturedYear}$$

$$\text{AgeScore} = 1 - \min(\text{Age} / 24, 1)$$

Devices 0–10 years old retain most of their value, while devices 20+ years old receive a score of 0. The reason for choosing 24 was to have an even 6 election cycles

2. OS Security Score

OS Score ranks operating systems by modernity and security posture.

Higher scores correspond to actively supported, secure OS versions, while outdated or unsupported OS versions score lower.

3. Certification Score

Certification Score reflects the VVSG certification status:

- VVSG 2.0 → 1.0
- VVSG 1.0 → 0.8
- Other or None → scaled accordingly.

4. Scan Score

For scanners only. Scan score normalizes scan rate performance:

$$\text{ScanScore} = \min(\text{ScanRate} / \text{MaxScanRateObserved}, 1)$$

Non-scanners bypass this metric, and their formula is renormalized.

5. Error Score

Error score reflects the likelihood of operational failure or malfunction. Where explicit data is unavailable, a default mid-value is used for all devices.

6. Reliability Score

Reliability Score blends device type reliability with Age Score:

$$\text{Reliability} = (\text{DeviceTypeReliability} + \text{AgeScore}) / 2$$

Device type reliability rankings (example):

$$\text{Scanner} = 0.9$$

$$\text{Ballot Marking Device} = 0.75$$

$$\text{DRE} = 0.6$$

$$\text{Lever/Punchcard} = 0.4$$

7. Security Risk Score

Security Risk Score is computed by detecting the presence of various high-severity and medium-severity risk tags.

Start at 1.0, then subtract:

$$-0.075 \text{ per high-severity risk}$$

$$-0.05 \text{ per medium-severity risk}$$

Clamped between 0.5 and 1.0:

$$\text{RiskScore} = \max(0.5, \min(1.0, 1 - 0.10 * H - 0.05 * M))$$

8. Final Weighted Quality Score

Final scores differ for scanners and non-scanners due to scan performance relevance.

Weights:

Certification (S): 0.25

Age (Q): 0.25

OS (R): 0.20

Error (V): 0.10

Security Risk (X): 0.10

Scan (T): 0.05

Reliability (U): 0.05

Scanner Formula

For scanner devices, all categories apply, including scan performance. The Final Quality Score for scanners is:

$$\text{Quality_scanner} = 0.25S + 0.25Q + 0.20R + 0.10V + 0.10X + 0.05T + 0.05U$$

This formula emphasizes certification, age, and operating system while still accounting for error behavior, concrete security risks, scan throughput, and overall reliability.

Non-Scanner Formula

Non-scanner devices (such as BMDs or certain electronic systems that do not scan ballots) do not use the scan metric T. To keep scores on a comparable 0–1 scale, the weights for the remaining terms are renormalized by dividing by the sum of their weights (0.95):

$$\text{Quality_non_scanner} = (0.25S + 0.25Q + 0.20R + 0.10V + 0.10X + 0.05U) / 0.95$$

This ensures that a non-scanner with perfect subscores can still achieve a Final Quality Score near 1.0, even though it does not have a scan-throughput component.

DRE Penalty

Direct Recording Electronic (DRE) systems are treated differently in the final calculation due to their architectural risks (software-only tabulation and, in many cases, lack of a voter-verifiable paper record). After computing the scanner or non-scanner Quality Score as above, any device whose type is identified as a DRE receives an additional multiplicative penalty of 0.8:

$$\text{Quality_DRE} = 0.9 \times \text{Quality_raw}$$

where Quality_raw is the score computed using the appropriate scanner or non-scanner formula. This ensures that, all else equal, DREs cannot score as highly as functionally similar devices that produce auditable paper records.

If a device is discontinued:

$$\text{Quality} = 0$$